

Open-Source & Open-Access Resource List

MIT 103 - Advanced Systems Analysis, Design, and Implementation

Partido State University - College of Engineering and Computational Sciences
First Semester, AY 2026-2027

40 curated resources	6 categories	18 weeks mapped	All free to access
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How to read this list

Every entry below is free to access. They are not all free to *reuse*, so each carries one of three badges. All titles and URLs are clickable in this PDF - click the blue link line under any entry to go straight to the download page. Section G maps the resources onto the 18-week instructional schedule in the OBE-aligned syllabus.

OPEN

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FREE

Free of charge to read and download, but NOT openly licensed. Do not re-host or remix; link to the source.

OSS

Open-source software released under an OSI-approved licence (GPL, MPL, Apache, EPL, MIT).

A. Core Open Textbooks - Information Systems, Analysis & Design

1. Foundations of Information Systems

OPEN

M. S. Raisinghani (Ed.), OpenStax / Rice University, 2025 • [CC BY-NC-SA 4.0](#)

Newest peer-reviewed IS textbook (published 5 Mar 2025). Covers IS fundamentals, data management, databases, systems development, security and emerging tech. Free PDF, web, and low-cost print. Instructor slides and test banks available.

↓ openstax.org/details/books/foundations-information-systems

2. Information Systems for Business and Beyond (2019 ed.)

OPEN

D. T. Bourgeois, J. L. Smith, S. Wang & J. Mortati, Saylor / Biola, 2019 • [CC BY-NC 4.0](#)

The most widely adopted IS open textbook. Ch. 10 (Information Systems Development) maps directly to SDLC, agile and OO methodologies in Week 2. Accessible PDF, EPUB and web versions.

↓ opentextbook.site/exports/ISBB-2019.pdf (direct PDF)

3. Information Systems for Business and Beyond - Canadian Edition

OPEN

T. Ramim et al., eCampusOntario Pressbooks • [CC BY-NC](#)

Localised adaptation with non-US cases; useful for contrasting organisational and regulatory contexts in the Week 3 systems-perspective discussion.

↓ ecampusontario.pressbooks.pub/informationssystemscdn

4. Business Information Systems: Design an App for That

OPEN

Open Textbook Library / Saylor Foundation • [CC BY](#)

Project-based book organised around the full SDLC - analyse, plan, design, build, deploy. Strong fit for the running System Proposal / Prototype portfolio output.

↓ open.umn.edu/opentextbooks/textbooks/business-information-systems-design-an-app-for-that

5. Database Design - 2nd Edition

OPEN

A. Watt & N. Eng, BCcampus Open Education • [CC BY 4.0](#)

142 pages on data models, ER modelling, functional dependency, normalisation and SQL. Includes ERD exercises and a full SQL lab with solutions - directly reusable for Week 8 (data dictionaries) and the Database Design deliverable.

↓ opentextbc.ca/dbdesign01

B. Software & Requirements Engineering

6. Requirements Engineering (open-source textbook)

OPEN

B. Regnell (Ed.), Lund University - continuously updated • [CC BY-SA 4.0](#)

LaTeX-source textbook plus ready-made lecture slide decks in the /slides and /lectures folders. Elicitation, specification, prioritisation and validation. Slides are reusable in your own deck for Weeks 4-5.

↓ github.com/bjornregnell/regeng-book

7. Handbook of Software Engineering Methods - 2nd Edition

OPEN

L. Letaw, Oregon State University, 2024 • [CC BY-NC 4.0](#)

Method-by-method handbook: user stories, personas, inclusive design (GenderMag), sprint planning, code review, usability testing. Practical scaffolding for lab activities.

↓ open.oregonstate.edu/setextbook

8. Guide to the Software Engineering Body of Knowledge (SWEBOK) v4.0

FREE

H. Washizaki (Ed.), IEEE Computer Society, 2024 (v4.0a rev. 2025) • [Free for personal & academic use](#)

The authoritative 18-knowledge-area reference. New KAs on Software Architecture, Operations and Security. Use KA 1 (Requirements), KA 2 (Architecture), KA 3 (Design), KA 8 (Quality) as graduate-level reading anchors.

↓ ieeecs-media.computer.org/media/education/swebok/swebok-v4.pdf (direct PDF)

9. Systems Engineering Body of Knowledge (SEBoK)

OPEN

INCOSE / IEEE / Stevens Institute - living wiki • [CC BY-NC-SA 3.0](#)

Free, citable wiki. Its ISO/IEC/IEEE 29148 article is the best legal substitute for the paywalled requirements standard.

↓ sebokwiki.org

10. Software Engineering: A Modern Approach

FREE

M. T. Valente, 2024 • *Free web version; all rights reserved*

Full web version free. 900+ instructor slides on Google Slides (request access via the site form). Excellent chapters on Processes, Requirements, Models (UML) and Architecture. Link to it - do not re-host.

↓ softengbook.org

11. Software Engineering: Standing on the Shoulders of Giants

OPEN

SWEBook project • *Creative Commons*

Open textbook with original prose, diagrams and exercises built around landmark SE papers - good seed material for graduate critique assignments (PO4 / d.1).

↓ swebook.org

C. Modelling Notations & Standards**12. OMG Unified Modeling Language (UML) 2.5.1 Specification**

FREE

Object Management Group, 2017 • *Free download for informational use*

The normative source for use case, class, sequence and activity diagrams. Cite this rather than a textbook redraw when students defend their models.

↓ omg.org/spec/UML/2.5.1/PDF (direct PDF)

13. OMG Business Process Model and Notation (BPMN) 2.0.2

FREE

Object Management Group / ISO 19510 • *Free download for informational use*

Use for enterprise process analysis in Weeks 3 and 7 as a modern complement to classic Gane-Sarson / Yourdon DFDs.

↓ omg.org/spec/BPMN/2.0.2/PDF (direct PDF)

14. The Scrum Guide (Nov 2020)

OPEN

K. Schwaber & J. Sutherland • *CC BY-SA 4.0*

13 pages, the definitive agile framework text. Free PDF in 30+ languages including Filipino.

↓ scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf (direct PDF)

15. Manifesto for Agile Software Development

FREE

Beck, Cockburn, Fowler et al., 2001 • *Free to reproduce in full*

Primary source for the Week 6 agile modelling discussion - four values, twelve principles.

↓ agilemanifesto.org

16. ISO/IEC 25010:2023 - Product Quality Model (SQuaRE)

FREE

ISO / IEC • *Read-only preview free; full text paid*

Nine quality characteristics (note: 'Usability' was renamed 'Interaction Capability' in the 2023 revision). Use the free Online Browsing Platform preview for the Week 18 quality-assurance criteria.

↓ iso.org/obp/ui (free read-only preview)

17. ISO/IEC/IEEE 29148 Requirements Specification Templates

FREE

ReqView (free templates) + SEBoK article • *Free templates; standard itself is paid*

Free downloadable SRS / StRS / SyRS templates conforming to 29148:2018 - a practical, no-cost stand-in for the paywalled standard.

↓ reqview.com/doc/iso-iec-ieee-29148-templates

D. Human-Computer Interaction, Interface & Output/Input Design**18. Introduction to Human-Computer Interaction**

OPEN

K. Hornbaek, P. O. Kristensson & A. Oulasvirta, Oxford University Press, 2025 • *CC BY-NC-ND 4.0 (open access)*

The newest comprehensive HCI textbook - 864 pages, 46 chapters, 200+ principles. Free chapter-level PDFs on Oxford Academic. Covers perception, cognition, user research, UI types, design, engineering and evaluation. Instructor slides available on request. Anchors Weeks 11-15.

↓ academic.oup.com/book/60808 (free chapter PDFs)

19. The Encyclopedia of Human-Computer Interaction, 2nd Ed.

FREE

M. Soegaard & R. F. Dam (Eds.), Interaction Design Foundation • [Free to read online](#)

4,000+ page reference with chapters by Norman, Shneiderman, Card and others. Good for assigning individual chapters as graduate seminar readings.

↓ interaction-design.org/literature/book/the-encyclopedia-of-human-computer-interaction-2nd-ed**20. 10 Usability Heuristics for User Interface Design**

FREE

J. Nielsen, Nielsen Norman Group (rev. 2024) • [Free article + downloadable poster](#)

The standard heuristic-evaluation instrument. Turn it into the rubric for the Week 15 peer-critique activity.

↓ nngroup.com/articles/ten-usability-heuristics**21. Web Content Accessibility Guidelines (WCAG) 2.2**

OPEN

W3C Recommendation, 5 October 2023 • [W3C Document Licence \(free\)](#)

Normative accessibility criteria for the input/output design weeks. Pair with the 'How to Meet WCAG' quick reference for a checkable student rubric.

↓ w3.org/TR/WCAG22**22. GOV.UK Design System & Service Manual**

OPEN

UK Government Digital Service • [MIT / Open Government Licence](#)

Production-grade, evidence-backed patterns for forms, error messages, question pages and data entry - exactly the Weeks 12-13 and 16-17 content, with research citations behind every pattern.

↓ design-system.service.gov.uk**23. U.S. Web Design System (USWDS)**

OSS

U.S. General Services Administration • [Public domain \(CC0\) / open source](#)

Accessible component library plus detailed form and validation guidance. Free code students can actually build their prototypes on.

↓ designsystem.digital.gov**24. Material Design 3**

FREE

Google • [Apache 2.0 \(code\)](#); [free guidelines](#)

Reference for dashboards, widgets, adaptive layouts and touch-screen/tablet interfaces - the Week 11 and Week 14 topics.

↓ m3.material.io**E. Open-Source Tools for Labs and Portfolio Work****25. draw.io / diagrams.net**

OSS

JGraph Ltd - Apache 2.0 • [Apache 2.0](#)

Free desktop and browser diagramming. Built-in DFD, ERD, UML and wireframe shape libraries. No account needed; files save locally - ideal for lab machines.

↓ drawio.com**26. PlantUML**

OSS

PlantUML project - GPL / LGPL / Apache • [GPL](#) / [LGPL](#) / [Apache](#)

Diagrams-as-code. Version-controllable UML - lets students track model evolution in Git and supports the PO6 collaboration outcomes.

↓ plantuml.com**27. Mermaid**

OSS

Mermaid.js - MIT • [MIT](#)

Markdown-embedded diagrams (flowchart, ER, sequence, Gantt). Renders natively in GitHub, GitLab and Notion - good for lightweight documentation.

↓ mermaid.js.org

28. Modelio

OSS

Modeliosoft - GPL v3 (Open Source Edition) • GPL v3

Full UML 2, BPMN 2 and ArchiMate modelling in one desktop tool. The strongest free option for formal, graduate-level enterprise modelling.

↓ modelio.org**29. Eclipse Papyrus**

OSS

Eclipse Foundation - EPL 2.0 • EPL 2.0

Industrial-strength UML/SysML modelling with model validation and code generation.

↓ eclipse.dev/papyrus**30. Archi (ArchiMate modelling tool)**

OSS

The Open Group community - MIT • MIT

Free ArchiMate tool for enterprise architecture views - supports the PO1 a.1 outcome on enterprise-level IT architectures.

↓ archimatetool.com**31. bpmn.io (Camunda Modeler)**

OSS

Camunda - bpmn-js MIT • MIT

Browser-based BPMN 2.0 process modeller for the enterprise process-analysis component.

↓ bpmn.io**32. Penpot**

OSS

Kaleidos - MPL 2.0 • MPL 2.0

The open-source alternative to Figma: wireframing, high-fidelity UI design and clickable prototypes. Self-hostable, so no student licence cost for Weeks 11-15.

↓ penpot.app**33. pgModeler / DBeaver**

OSS

pgModeler (GPL v3) and DBeaver Community (Apache 2.0) • GPL v3 / Apache 2.0

ER design, forward/reverse engineering and SQL work for the Week 8 database-design deliverable.

↓ pgmodeler.io**34. Taiga**

OSS

Kaleidos - AGPL v3 • AGPL v3

Open-source Scrum/Kanban project management. Use it so students practise sprint planning and backlog grooming rather than only reading about it.

↓ taiga.io**F. Courseware, Repositories & Ongoing Sources****35. MIT OpenCourseWare**

OPEN

Massachusetts Institute of Technology • CC BY-NC-SA 4.0

Full lecture notes, assignments and exams. Relevant: ESD.71 Engineering Systems Analysis for Design; 2.882 System Design and Analysis; 6.170 Software Studio.

↓ ocw.mit.edu**36. Open Textbook Library - Information Systems**

OPEN

University of Minnesota Open Education Network • Varies (all open licences)

Peer-reviewed catalogue of IS open textbooks with faculty reviews - the first place to check when refreshing this list each semester.

↓ open.umn.edu/opentextbooks/subjects/information-systems**37. BCCampus Open Collection**

OPEN

BCCampus Open Education • Varies (all open licences)

Downloadable PDF, EPUB and editable Pressbooks source for every title - so you can legally adapt chapters into PARSU-branded handouts.

↓ collection.bccampus.ca

38. Saylor Academy - Computer Science courses

[OPEN](#)

Saylor Academy • [CC BY 3.0](#)

Free, self-paced courses with assessments. CS302 Software Engineering and CS403 Modern Database Systems align with several weeks of this syllabus.

↓ learn.saylor.org

39. arXiv cs.SE - Software Engineering preprints

[OPEN](#)

Cornell University • [Author-selected open licences](#)

Current research feed. Assign recent preprints for the PO4 critical-evaluation outcome and the Chapters 1-5 research-format final output.

↓ arxiv.org/list/cs.SE/recent

40. DOAJ - Directory of Open Access Journals

[OPEN](#)

DOAJ • [Open access](#)

Vetted open-access journals in information systems and software engineering for the students' literature review.

↓ doaj.org

G. Mapping to the 18-Week Instructional Schedule

Suggested primary readings and tools per week, following the OBE-aligned course syllabus (PSU-SYL-MIT101, Rev. 03).

Wk	Instructional content	Recommended open resources
1	Orientation: VMGO, course content, grading	Course syllabus; PARSU Student Handbook; OpenStax Foundations of IS (Ch. 1 preview)
2	Systems, roles, and development methodologies (SDLC, Agile, OO)	ISBB 2019 Ch. 10; SWEBOK v4 KA 1-3; Scrum Guide 2020; Agile Manifesto
3	Understanding & modelling organizational systems	OpenStax Foundations of IS; SEBoK; BPMN 2.0.2; Archi; draw.io
4	Information gathering: interactive methods	Regnell, Requirements Engineering (elicitation ch.); Letaw, Handbook of SE Methods; Intro to HCI Part III (User Research)
5	Information gathering: unobtrusive methods, STROBE	Intro to HCI Ch. 14 (Unobtrusive Research); Letaw Handbook; ReqView 29148 templates
6	Agile modeling and prototyping	Scrum Guide 2020; Agile Manifesto; Valente Ch. 2; Penpot; Taiga
7	Using data flow diagrams (logical & physical)	draw.io DFD library; BPMN 2.0.2; Business Information Systems: Design an App for That
8	Analyzing systems using data dictionaries	Watt & Eng, Database Design 2e; pgModeler / DBeaver; OpenStax Foundations of IS (data ch.)
9	MIDTERM EXAMINATION	-
10	Process specifications and structured decisions	SWEBOK v4 KA 3 (Design); Valente Ch. 4 (Models); PlantUML activity diagrams
11	Output design: displays, dashboards, widgets, web output	Intro to HCI Parts V & VI; Material Design 3; GOV.UK Design System; WCAG 2.2; Penpot
12	Input design: functional and engaging input displays	GOV.UK Design System (forms & question pages); USWDS form guidance; WCAG 2.2
13	Web, intranet and tablet input design	USWDS; Material Design 3 (adaptive layouts); GOV.UK patterns; Penpot
14	Introduction to HCI; user-centred interaction; touch interfaces	Intro to HCI (Hornbaek et al.) Parts I-IV; ISO 9241-210 overview via SEBoK; NN/g heuristics
15	Types of user interfaces; onscreen dialog; applying HCI	Intro to HCI Parts V & VIII; NN/g 10 Heuristics (evaluation rubric); Penpot prototypes
16	Data entry procedures and effective coding	GOV.UK Service Manual; USWDS; Watt & Eng Database Design 2e (constraints)
17	Data validation, verification and data quality	WCAG 2.2 error-prevention criteria; USWDS validation patterns; ISO/IEC 25010:2023
18	Quality assurance, implementation, SOA and cloud	SWEBOK v4 KA 8 & new Operations/Security KAs; ISO/IEC 25010:2023; Valente Ch. 8-10 (Testing, Refactoring, DevOps)

H. Licensing Notes and Maintenance

- Verify licence terms before re-hosting. Items badged FREE (SWEBOOK, OMG specs, ISO previews, softengbook.org, IxDF) may be downloaded and read at no cost but are NOT openly licensed - link to the official source instead of uploading copies to Google Classroom.
- Items badged OPEN may be legally copied, excerpted and adapted into PARSU-branded handouts, provided you attribute the author and keep the same licence where ShareAlike (SA) applies.
- NonCommercial (NC) licences permit classroom use at a public university; they do not permit resale or inclusion in paid training products.
- ISO and IEEE standards themselves are paid. The free substitutes listed here (SEBoK articles, ReqView templates, ISO Online Browsing Platform previews) are sufficient for teaching, but cite the standard number in student deliverables.
- Recommended review cycle: re-check this list every semester. OpenStax, BCcampus and the Open Textbook Library are the three sources most likely to add new titles in this subject area.

Suggested attribution line for adapted materials

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